Assignment No. 05: Basic Statistics (Measures, Moments, Expectation, Skewness)

Blue = data from question; black = formulas. Σ in column K. Class intervals treated as continuous; moments via binomial shift formulas.

Q1. Mean, median and mode of the frequency distribution (discrete)											Σ
x	1	2	3	4	5	6	7	8	9	10	
f	4	7	8	10	6	6	4	2	2	1	50
f·x	4	14	24	40	30	36	28	16	18	10	220
cum. f	4	11	19	29	35	41	45	47	49	50	
Results	n	50	Mean	4.4000	Median	4.	Mode	4.			

Q2. Insurance claims (thousand ₹): mean, median and mode (continuous classes)											Σ
Lower L	1	3	5	7	9	11					
Upper U	3	5	7	9	11	13					
f	6	47	75	46	18	8					200
x (mid)	2.0	4.0	6.0	8.0	10.0	12.0					
f·x	12.0	188.0	450.0	368.0	180.0	96.0					1294.0
cum. f	6	53	128	174	192	200					
Results	n	200	Mean	6.4700	Median	6.2533	Mode	5.9825			

Q3. Mean deviation and standard deviation of the data											Σ
Size x	6	7	8	9	10	11	12				
f	3	6	9	13	8	5	4				48
f·x	18	42	72	117	80	55	48				432
|x − mean|	3.00	2.00	1.00	0.00	1.00	2.00	3.00				
f·|x−mean|	9.00	12.00	9.00	0.00	8.00	10.00	12.00				60.00
f·x²	108	294	576	1053	800	605	576				4012
Results	n	48	Mean	9.0000	M.D.	1.2500	Variance	2.5833	S.D.	1.6073	

Q4. Quartile deviation and mean deviation (about mean) of the distribution											Σ
Lower L	0	10	20	30	40	50	60	70			
Upper U	10	20	30	40	50	60	70	80			
f	2	5	7	13	21	16	8	3			75
x (mid)	5	15	25	35	45	55	65	75			
f·x	10	75	175	455	945	880	520	225			3285
cum. f	2	7	14	27	48	64	72	75			
|x − mean|	38.80	28.80	18.80	8.80	1.20	11.20	21.20	31.20			
f·|x−mean|	77.60	144.00	131.60	114.40	25.20	179.20	169.60	93.60			935.20
Results	n	75	Mean	43.8000	Q1	33.6538	Q3	55.1563	Q.D.	10.7512	
Mean dev.	M.D.	12.4693									

Q5. First four moments about x = 4 are −1.5, 17, −30, 108: moments about mean, β1, β2, and about x = 2					Σ
Points	a (given)	4	point	2	
Moment r	1	2	3	4	
About x = 4	-1.50	17.00	-30.00	108.00	
About mean μr	0.0000	14.7500	39.7500	142.3125	
About x = 2	0.5000	15.0000	62.0000	244.0000	
Results	β1	0.4924	β2	0.6541	Positively skewed; platykurtic (β2<3)

Q6. f(x) = 4x(9−x²)/81, 0≤x≤3: first four moments (a) about origin, (b) about mean						Σ
Moment r	0	1	2	3	4	
About origin	1.0000	1.6000	3.0000	6.1714	13.5000	r = 0 gives total probability = 1 (check)
About mean		0.0000	0.4400	-0.0366	0.4221	

Q7. Moments about a = 40.5 and then about the mean (class marks 31.5, 34.5, … used as mid-values, h = 3)											Σ
Constants	a	40.5	h	3	n	100					
Class	30-32.9	33-35.9	36-38.9	39-41.9	42-44.9	45-47.9					
x (mid)	31.5	34.5	37.5	40.5	43.5	46.5					
f	2	4	26	47	15	6					100
d=(x−a)/h	-3	-2	-1	0	1	2					
f·d^1	-6	-8	-26	0	15	12					-13
f·d^2	18	16	26	0	15	24					99
f·d^3	-54	-32	-26	0	15	48					-49
f·d^4	162	64	26	0	15	96					363
Moment r	1	2	3	4							
About a = 40.5	-0.3900	8.9100	-13.2300	294.0300							
About mean	0.0000	8.7579	-2.9239	281.4531							

Q8. Bag with 3 white and 5 black balls, 2 drawn; ₹50 per white, ₹20 per black: expectation											Σ
Inputs	White	3	Black	5	₹/white	50	₹/black	20			
Whites drawn	2	1	0								
Ways	3	15	10								28
P(x)	0.1071	0.5357	0.3571								1.0000
Gain X (₹)	100	70	40								
P·X	10.71	37.50	14.29								62.50
Result	E(X) ₹	62.50	Check ₹	62.50							

Q9. f(x) = x/2, 0<x<2: E(X), V(X), E(3X² − 2X) [E(X^r) = ∫₀² x^r·x/2 dx]						Σ
r	0	1	2			
E(X^r)	1.0000	1.3333	2.0000	r = 0 → total probability = 1 (check)		
Results	E(X)	1.3333	V(X)	0.2222	E(3X²−2X)	3.3333

Q10. n = 10, Σx = 452, Σx² = 24270, mode = 43.7: Karl Pearson's coefficient of skewness								Σ
Given	n	10	Σx	452	Σx²	24270	Mode	43.7
Results	Mean	45.2000	Variance	383.9600	S.D.	19.5949	Sk	0.0766